

Gemini Model Parameters – Detailed Explanation

Model Selection & Basic Parameters

i. **model_name**

Purpose: Choose which Gemini model to use

- **gemini-pro:** General text generation, conversations, reasoning
- **gemini-pro-vision:** Multimodal tasks (text + images)
- **gemini-1.5-pro:** Latest with 1M token context, better reasoning
- **gemini-1.5-flash:** Fast, lightweight, cost-effective
- **gemini-ultra:** Most capable (when available)

ii. **temperature**

Purpose: Controls randomness/creativity

- **0.0:** Deterministic, factual, consistent
- **0.3–0.7:** Balanced
- **0.8–1.0:** Highly creative

iii. **top_p**

Purpose: Nucleus sampling – controls diversity

iv. **top_k**

Purpose: Limits sampling to top K most probable tokens

Response Control Parameters

i. **n**

Purpose: Number of responses to generate

ii. **max_output_tokens**

Purpose: Maximum length of response

iii. **stop**

Purpose: Stop generation when specific strings appear

API & Connection Parameters

- i. **timeout**
Purpose: API call timeout in seconds
- ii. **max_retries**
Purpose: Number of retry attempts on failure
- iii. **google_api_key**
Purpose: Google AI API key (keep secure)
- iv. **google_api_base**
Purpose: Custom API endpoint URL
- v. **client_options**
Purpose: Advanced client configuration
- vi. **transport**
Purpose: HTTP transport method
- vii. **client**
Purpose: Provide custom client instance

Safety & Functionality

- i. **safety_settings**
Purpose: Control content filtering
- ii. **tools**
Purpose: Enable function calling
- iii. **tool_config**
Purpose: Configure how tools are used

Message & Streaming

- i. **convert_system_message_to_human**
Purpose: Handle system messages
- ii. **streaming**
Purpose: Stream responses token by token
Practical Examples and Recommendations are included for real-world usage.